

# Run a Go and No Go Gauge Battery SOP

One episode gauges ten parts. It runs from ten parts standing in a numbered rack, two bare gauge posts, two empty bins and a blank tally card, to all ten parts gauged two ways each, ten tally strokes and three flag marks on the card, seven parts in the pass bin, three parts in the reject bin and every tool back on its spot.

One part is finished before the next one is started. The order for each part never changes: **take the part out of its slot, run it on the GO post, run it on the NO GO post, mark the card, drop the part in its bin.** No part is gauged until both posts have been checked with the master ring, and no part goes in a bin until its stroke is on the card.

**One arm does all of it.** The same gripper takes every part out of the rack, lowers it onto each post, lifts it off again, works the marker and drops the part in its bin. Nothing in this task has to be held, because nothing moves on its own: the rack, the gauge block, both bins and the tally card all grip the table, and a part left on a post stands there on its own. Only one arm works in this cell. Do not bring a second arm into the task.

Every spot sits inside the arm's comfortable reach, so the arm never has to stretch or fold in on itself. Only one part is held at a time, no grasp is changed while a part is off the table, and no part is ever carried over a bin it is not going into. A good part leaves the gauge block to the right. A reject leaves the gauge block to the left.

The gauges do the deciding, not the eye. A part is good only if it drops all the way down on the GO post and stops high on the NO GO post. Nothing is measured, no number is read, and no part is compared against another part.

The card is written in the plainest words the job allows. A part is not marked out of tolerance or oversize: it is marked **TOO BIG** or **TOO SMALL**, and a part that is neither is marked **GOOD**. Anyone who can read those three lines can read the card, and the flag boxes say which part was the bad one.

Each tool goes back to its own spot before the next step starts. The counts are fixed: ten parts, one master check, two gauge runs per part, ten tally strokes, three flag marks, seven parts in the pass bin and three parts in the reject bin. There is no second master check and no second run of the same gauge on the same part. A part that lands off the table is left exactly where it falls, tagged, and the episode continues. Nothing is ever picked up off the floor on camera.

## Setup

Complete both checklists before starting an episode.

### Cell configuration

- **Environment camera:** 90 mm.

- **cell\_type:** single\_arm

## **Hardware checklist**

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the parts rack, the gauge block with both posts, the master ring spot, both bins, the tally card and the marker holder.
3. The foot of each post stays in frame the whole time, so a reviewer can see whether a part has come down onto the shoulder or is standing high above it.
4. All three rows of the tally card and all ten flag boxes stay in frame and readable the whole time.
5. The arm is at home with its gripper open.
6. The tabletop is dry and clear of anything but the items in the Materials checklist.
7. The arm reaches every spot on the table, all ten rack slots, both posts, the master ring spot, both bins, the tally card and the marker holder, without extending to a joint limit or folding in on itself. Bring the spots forward or closer together until this is true.
8. The arm reaches every one of the ten rack slots straight on, so a part can be lifted straight up out of any slot without leaning it over.
9. The arm comes down onto each post straight from above, so a part can be lowered onto the post without being tilted or swung in.
10. The paths from each rack slot to the GO post, from the GO post to the NO GO post, from the NO GO post to each bin, and from the marker holder to the card are unobstructed. Carry one part along each path by hand during setup: it must clear the top of both posts and the rim of every bin it passes.

## **Materials checklist**

1. One parts rack stands at the back of the table, straddling the midline, with ten upright slots in two rows of five, numbered 1 to 5 across the back row and 6 to 10 across the front row, left to right. The numbers face the camera.
2. The rack grips the table. Push it by hand during setup: if it slides, turns or lifts, weight it or replace it. Nothing pins the rack during the episode.
3. Each slot holds one part standing upright, bore up, with enough clearance that the part lifts straight out without rubbing.
4. Ten parts, and no more. Each part is a plain round ring with a straight bore through the middle, flat faces top and bottom, no burrs and no markings on it. Every part is the same on the outside, so the outside tells a reviewer nothing and only the gauges do.
5. Each part is tall enough for the gripper to close on its outside squarely, and short enough to sit well below the top of either post when it is standing on the shoulder.

6. The ten parts are the same ten in every episode: seven good parts, two parts with a bore that is too big, and one part with a bore that is too small.
7. The three bad parts are bad by a plain amount. A too big part drops all the way down on the NO GO post with no rocking, and a too small part stops well clear above the GO post shoulder with daylight anyone can see. A part that only just fails is not used.
8. One gauge block sits at the middle of the table, straddling the midline. It is heavy and it grips the table.
9. Two gauge posts stand upright on the block, the GO post at the back of the block and the NO GO post in front of it, both on the midline with a clear gap between them. Each post is labelled GO or NO GO on the block facing the camera, and the same post is the GO post in every episode.
10. The GO post is the smaller of the two. A good part's bore drops over it freely. The NO GO post is the larger. A good part's bore will not pass over it.
11. Each post stands on a flat shoulder at its foot, wide enough to carry a part squarely and clear enough that a part sitting on it is plainly down rather than standing high.
12. Both posts are fixed in the block. Twist each one by hand during setup: if a post turns, rocks or lifts, tighten it or replace the block. No post is ever adjusted during the episode.
13. Both posts and every part are clean and dry. A trace of oil, grit or swarf on a post or in a bore makes a good part read as a bad one.
14. One master ring lies on its own rest on the gauge block, at the back left corner and clear of both posts. It is a known good ring, marked so it cannot be mistaken for a part, and it is never binned and never tallied.
15. Two bins stand in their holders on the table: the pass bin right of the gauge block and the reject bin left of the gauge block. They are empty and dry, they are clearly different in colour, each is labelled PASS or REJECT facing the camera, and the same bin is the pass bin in every episode.
16. Each bin is low enough and wide enough that a part can be dropped in from just above the rim, and each one sits in its holder so it cannot tip or slide when a part lands in it.
17. One tally card lies flat at the front centre on the midline. It is a wipe clean card and it starts the episode blank.
18. The card carries three tally rows stacked down it, labelled down the left in plain words: GOOD on the top row, TOO BIG on the middle row, TOO SMALL on the bottom row. Each row holds one long open tally box, wide enough for ten strokes standing apart.
19. Under the three rows the card carries one flag strip: ten small boxes in a line, numbered 1 to 10 left to right, matching the ten rack slots, under the heading REJECTS. Every box is big enough to hold one X with room to spare, and the boxes do not touch each other.
20. The words on the card are printed in full and in plain English. No code, no number range and no gauge language appears anywhere on it.

21. The card grips the table. Push it by hand during setup: if it slides, tape it down or put it in a holder.
22. One dry erase marker stands in its holder on the card's right edge, cap off, with enough ink to draw thirteen marks.

## Workspace layout

Three rows: the parts at the back, the gauge block in the middle with a bin on each side of it, and the tally card at the front. Every part travels forward out of the rack to the GO post, forward again to the NO GO post, then sideways off the block into a bin.

Back row:

1. Parts rack, back centre on the midline, ten upright slots in two rows of five, slots 1 to 5 across the back row and slots 6 to 10 across the front row, left to right, with the numbers facing the camera.

Middle row, left to right:

2. Reject bin spot, left of the gauge block, the bin standing in its holder. A reject leaves the gauge block to the left.
3. Gauge block, centre, straddling the midline, carrying the GO post at its back and the NO GO post in front of it, both on the midline, with the master ring on its rest at the block's back left corner. Both posts are bare at the start of the episode and bare again at the end.
4. Pass bin spot, right of the gauge block, the bin standing in its holder. A good part leaves the gauge block to the right.

Front row:

5. Tally card, front centre on the midline, its three rows and its flag strip facing the camera.
6. Marker holder, on the tally card's right edge.

A clear gap of at least a gripper's width sits between the two posts, between the master ring rest and both posts, and between the gauge block and each bin, so a part comes down onto one post without touching the other, and drops into a bin straight from above without the gripper touching the block. No carry passes over the bin the part is not going into, and no carry passes over the top of a post it is not going onto.

## Vocabulary

- **Arm assignment:** one gripper does everything: every part, the master ring, every mark. This is a single arm task and no second arm takes any part in it.
- **Tool home:** the spot a tool returns to and sits on before the next step starts: the marker holder and the master ring rest. A tool is never left on a post, on the card, in a bin or in the gripper between steps.

- **Release point:** the gripper opens over a post when a part has come to rest on it, just above the rim of a bin when a part is being dropped in, or over a tool home. Never over the wrong bin and never over open tabletop.
- **Stable placement:** the object stands still after release, square on its surface, and does not rock, tip, slide or lean on anything.
- **Part:** one plain round ring with a bore through the middle. There are ten in every episode, one to a rack slot.
- **Bore:** the hole through the middle of a part. It is the only thing being gauged, and it is never touched by the gripper.
- **GO post:** the smaller post, at the back of the gauge block. Every part is run on it first. A good bore drops all the way down over it.
- **NO GO post:** the larger post, in front of the GO post. Every part is run on it second. A good bore will not pass over it.
- **Shoulder:** the flat foot each post stands on. It is what a part comes to rest on when it drops all the way down.
- **Down:** the part has come all the way down the post and its bottom face rests flat on the shoulder, with no daylight under it anywhere.
- **Standing high:** the part has stopped on the post with its bottom face clear above the shoulder, daylight all the way round under it, and it stays there when the gripper opens.
- **Settled result:** the part has been let go, the gripper has come away, and the part has stayed where it stopped for 2 seconds with nothing touching it or the block.
- **Master ring:** the known good ring on its rest on the block. It is run on both posts once at the start of the episode to prove the gauges, then it goes back to its rest. It is never tallied, never flagged and never binned.
- **Master check:** the one time in the episode the master ring is run on the two posts. It happens once, before the first part.
- **Good part:** a part that went down on the GO post and stood high on the NO GO post. It is tallied GOOD and it goes in the pass bin.
- **Too big:** a part whose bore went down on the NO GO post. The hole is bigger than it should be. It is tallied TOO BIG.
- **Too small:** a part that stood high on the GO post. The hole is smaller than it should be. It is tallied TOO SMALL.
- **Reject:** any part tallied TOO BIG or TOO SMALL. It is flagged on the strip and it goes in the reject bin.
- **Tally stroke:** one straight upright stroke inside a tally box, standing on its own and not touching the stroke before it. One part draws one stroke, in one row.
- **Flag box:** one of the ten numbered boxes on the flag strip. It carries an X only when the part with that slot number is a reject, and it stays blank for a good part.

- **Marked box:** a box on the card holding its mark, fully inside its lines and not touching the next box.
- **One part at a time:** the gripper takes one part out of the rack, gauges it, marks it and bins it before it touches the next part.
- **Bare post:** nothing on the post and nothing on its shoulder.
- **Spill:** a part outside the rack, the posts and both bins. It is left where it falls and cleared in the reset, never on camera.
- **Quarter turn:** the most any object is tilted in the gripper. Every part travels bore up and level, and nothing is ever turned over in the air.

## Steps

### Step 1: Check both gauges with the master ring

**Goal:** the master ring down on the GO post, standing high on the NO GO post, and back on its rest, with both posts bare again.

The master check happens once, before the first part. If a post is worn, dirty or wrong, every result in the episode is wrong, so the gauges are proved before anything is gauged.

#### 1.1 Take the master ring

- **Check:** both posts are bare, with nothing on either shoulder.
- With the **gripper**, close on the outside of the master ring square, and lift it straight up off its rest until it is clear of the block.
- Keep it level, bore up, the whole way.

**Check:** the master ring hangs level in the gripper and does not slip. If it hangs tilted, set it back on its rest and take it again. Make no more than two corrective takes.

#### 1.2 Run the master ring on the GO post

- Carry it over the GO post, line the bore up with the top of the post, and lower it straight down slowly until it stops on its own.
- Stop there. Do not push it, twist it or press it down.
- Open the gripper, take it right away from the ring, the post and the block, and hold still.
- Wait 2 seconds with nothing touching it.

**Check:** the master ring is down, resting flat on the GO shoulder with no daylight under it. If it stands high, end the episode and clean or replace the gauge during reset.

#### 1.3 Run the master ring on the NO GO post

- With the **gripper**, close on the outside of the master ring square, and lift it straight up off the GO post until it is clear of the post top.

- Carry it forward to the NO GO post, line the bore up with the top of the post, and lower it straight down slowly until it stops on its own.
- Open the gripper, take it right away, hold still, and wait 2 seconds.

**Check:** the master ring is standing high on the NO GO post, with daylight all the way round under it. If it goes down, end the episode and check or replace the gauge during reset.

#### **1.4 Park the master ring**

- With the **gripper**, close on the outside of the master ring square, lift it straight up off the post, carry it back to its rest, lower it until it takes its own weight, release, and lift clear.

**Check:** the master ring sits flat on its rest, both posts are bare, and neither post has turned or lifted in the block. One master check, and one only. It is not run again between parts.

### **Step 2: Take the part from its slot**

**Goal:** one part out of its slot, hanging level in the gripper, and its slot empty.

Parts are worked in slot order, slot 1 first and slot 10 last. The rack and the block both grip the table, so nothing holds either of them while the part is moved.

#### **2.1 Lift the part straight out**

- With the **gripper**, close on the outside of the part square, well clear of the bore, and lift it straight up out of the slot until it is clear of the rack.
- Keep it level, bore up. Do not lean it over on the way out.
- The gripper never goes into the bore and never touches the inside of a part.

**Check:** the part hangs level in the gripper and does not slip. If it hangs tilted, stand it back in its slot and take it again. Make no more than two corrective takes. If it still will not hang level, end the episode and correct the part during reset.

### **Step 3: Run the part on the GO post**

**Goal:** the part resting where its own bore stopped it on the GO post, let go of, and settled for 2 seconds in frame.

The part is lowered and then released. It is never held while the result is being seen, because a held part can be steadied into looking down when it is not.

#### **3.1 Lower the part onto the GO post**

- Carry the part forward to the GO post, keeping it level and clear of the NO GO post, and bring it over the top of the post.
- Line the bore up with the post and lower it straight down slowly until it stops on its own.
- Stop the moment it stops. Do not push it down, twist it, screw it on or tap it onto the post.

**Check:** the part sits square on the post, level, touching nothing but the post and its shoulder, and the block has not moved.

### 3.2 Let go and look

- Open the gripper and take it right away from the part, the post and the block.
- Hold still and wait 2 seconds with nothing touching the part.
- **If the part is down**, resting flat on the shoulder with no daylight under it, the GO gauge is passed.
- **If the part is standing high**, with daylight all the way round under it, the part is **TOO SMALL**.

The camera records the result. It is not written down anywhere but the card, and it is written down in Step 5.

**Check:** the foot of the post is readable in frame and the part has stayed where it stopped. If the part slides down or settles after the 2 seconds, end the episode and let review flag it.

### Step 4: Run the part on the NO GO post

**Goal:** the same part resting where its own bore stopped it on the NO GO post, let go of, and settled for 2 seconds in frame.

Both gauges are run on every part, every time. A part that has already stood high on the GO post is still run on the NO GO post, so every part in the episode is gauged the same way.

#### 4.1 Lift the part off the GO post

- With the **gripper**, close on the outside of the part square, and lift it straight up off the GO post until it is clear of the post top.
- Keep it level. Do not tilt it off the post or drag it against the post on the way up.

**Check:** the part is clear of the GO post, the post is bare, and the block has not moved.

#### 4.2 Lower the part onto the NO GO post

- Carry the part forward to the NO GO post, keeping it level, and bring it over the top of the post.
- Line the bore up with the post and lower it straight down slowly until it stops on its own.
- Stop the moment it stops. Do not push, twist, screw or tap it on.

**Check:** the part sits square on the post, level, and touching nothing but the post.

#### 4.3 Let go and look

- Open the gripper and take it right away from the part, the post and the block.
- Hold still and wait 2 seconds with nothing touching the part.

- **If the part is standing high**, with daylight all the way round under it, the NO GO gauge is passed.
- **If the part is down** on the shoulder, the part is **TOO BIG**.
- Leave the part standing on the NO GO post. It stays there while the card is marked.

**Check:** the foot of the post is readable in frame, the part has stayed where it stopped, and the gripper is clear of the block.

## Step 5: Mark the card

**Goal:** one tally stroke in the one row the two gauges say, one X in the flag box if the part is a reject, and the marker back in its holder.

The part is still standing on the NO GO post while the card is marked, so what is being written down is still there to be seen.

### 5.1 Draw one tally stroke

- With the **gripper**, close on the marker barrel, lift it straight up out of its holder, and bring the tip over the card.
- **If the part went down on the GO post and stood high on the NO GO post**, draw one stroke in the **GOOD** row.
- **If the part went down on the NO GO post**, draw one stroke in the **TOO BIG** row.
- **If the part stood high on the GO post**, draw one stroke in the **TOO SMALL** row.
- Draw the stroke upright, fully inside the tally box, standing clear of the stroke before it.

One stroke per part, in one row only. Every part draws a stroke, good or not.

**Check:** exactly one new stroke is on the card, it is in the right row, it is inside its box, and it does not touch the stroke beside it.

### 5.2 Flag a reject

- **If the stroke went in the GOOD row**, the part is not a reject. Leave the flag strip alone and go to Step 5.3.
- **If the stroke went in the TOO BIG row or the TOO SMALL row**, go to the flag strip, find the box with the part's slot number on it, and draw one X in it, fully inside its lines.
- One X, in one box, on the number the part came out of.

The row says what is wrong with it in plain words and the flag box says which part it was. Nothing else is written on the card.

**Check:** the flag strip carries one X for every reject so far and no X for any good part, each one inside its own box.

### 5.3 Park the marker

- With the **gripper**, carry the marker back to its holder, lower it in until it stands on its own, release, and lift clear.

**Check:** the marker stands in its holder and the card carries one stroke for every part gauged so far. A wrong mark is not rubbed out and not drawn over. Leave it and let review flag the episode.

### Step 6: Put the part in its bin

**Goal:** the part in the bin its row says, and both posts bare again.

Both bins stand in their holders, so a part can be dropped into either one without anything holding it. Neither bin is ever lifted.

#### 6.1 Lift the part off the NO GO post

- With the **gripper**, close on the outside of the part square, and lift it straight up off the post until it is clear of the post top.

**Check:** the part is clear of the post and the gripper has not knocked the block.

#### 6.2 Drop it in its bin

- **If the part was tallied GOOD**, carry it off the gauge block **to the right** to the pass bin.
- **If the part was tallied TOO BIG or TOO SMALL**, carry it off the gauge block **to the left** to the reject bin.
- Bring it over the bin mouth, low to the rim, stop, and open the gripper so it drops straight in.
- A part never passes over the bin it is not going into.

**Check:** both posts are bare and the part is in the bin its row says. If a part has gone into the wrong bin, leave it there and let review flag the episode. Parts are never taken back out of a bin.

### Step 7: Do the next part

- Come back empty to the rack and take the part from the next slot in order.
- Run Steps 2 to 6 again for that part, flagging it on the box with its slot number if it is a reject.
- Ten parts, worked one at a time from slot 1 to slot 10. A part is finished and in a bin before the next part is touched.
- The master check is not run again. The gauges are proved once per episode.

**Check:** after each part, the rack has one fewer part in it, both posts are bare, one more stroke is on the card, and no part is gauged twice or skipped.

### Step 8: End the episode

- Confirm the rack: all ten slots empty.

- Confirm the posts: both bare, with the master ring on its rest.
- Confirm the bins: seven parts in the pass bin, three parts in the reject bin, both bins upright in their holders.
- Confirm the card: ten strokes in all, seven in the GOOD row, two in the TOO BIG row and one in the TOO SMALL row, each stroke inside its box and standing clear of the next.
- Confirm the flag strip: three boxes marked with an X, one for each reject, and the other seven blank.
- Confirm the tools: the marker upright in its holder, the master ring flat on its rest.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return the arm home with its gripper open, then stop recording.

## **After the episode: reset the workspace**

This reset is not recorded.

1. Take the parts out of both bins and stand the bins back in their holders, empty and dry.
2. Count the parts back into the rack: exactly ten, seven good, two too big and one too small. Replace any part that has been chipped, dented or burred so the ten are the same ten next time.
3. Shuffle the ten parts into the ten slots so they do not sit in the same order twice in a row, then stand each one upright, bore up.
4. Pick up any part that fell off a post or off the table and put it back in the rack. Replace it if it is damaged.
5. Wipe the tally card clean and dry it, so all three rows and all ten flag boxes start blank.
6. Wipe both posts, both shoulders, the master ring and every bore clean and dry. Grit or oil in a bore turns a good part into a reject.
7. Run the master ring on both posts by hand: it has to go down on the GO post and stand high on the NO GO post. Replace a post that fails this.
8. Set the master ring back flat on its rest at the back left corner of the block.
9. Twist both posts by hand and confirm neither turns, rocks nor lifts in the block.
10. Stand the marker in its holder with the cap off. Replace a marker that is running dry.
11. Confirm the rack, the gauge block, both bins and the tally card all still grip the table.
12. Confirm the card is still printed in full plain words, with GOOD, TOO BIG and TOO SMALL readable and the ten flag boxes still numbered 1 to 10.
13. Run both Setup checklists again.

# SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

## How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

## Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

## Violations

### Violation: Wrong order

- **Visible cue:** a part is gauged before the master check, the NO GO post is run before the GO post, a part goes in a bin before its stroke is on the card, a flag mark is drawn before the stroke, a new part is taken while the last one is still on a post, or a step is skipped.
- **SOP rule broken:** Steps 1 to 7, prove the gauges once, then for each part take it, run GO, run NO GO, tally it, flag it if it is a reject, bin it, in that order.
- **Coaching note:** gauges first, then one part all the way through. The stroke goes on the card while the part is still on the post.

### Violation: A second arm used

- **Visible cue:** a second arm enters the work area, closes its gripper, touches an object, steadies the rack, the block, a post, a bin or the card, or moves in front of the posts or the card. Any object taken or carried by a second gripper is a violation on its own.
- **SOP rule broken:** Steps 1 to 8, this is a single arm task: one gripper does all of it and nothing else in the cell touches the task.
- **Coaching note:** one arm only. If a second arm has worked, the episode is a bimanual episode wearing the wrong label.

### Violation: Object held or steadied

- **Visible cue:** the gripper presses on the rack, the gauge block, a post, a bin or the card to keep it still while working, or holds the part on a post while the result is being seen, instead of letting the block and the post do it.
- **SOP rule broken:** Steps 1 to 6, every object grips the table or stands in its holder, and nothing on this station is held.

- **Coaching note:** if something needs holding, the fixture is wrong. Fix the station, do not add a hand.

#### **Violation: Master check skipped, or run more than once**

- **Visible cue:** the first part goes on a post with no master check on camera, the master ring is run on only one of the two posts, the master result is called before it has settled for 2 seconds, or the master ring goes back on a post between parts.
- **SOP rule broken:** Step 1, one master check on both posts, before the first part, and one only.
- **Coaching note:** prove the gauges once, at the start. A master run in the middle means the results before it are being doubted.

#### **Violation: Part forced onto a post**

- **Visible cue:** the part keeps going down after it has stopped, is pushed, pressed, twisted, screwed or tapped onto a post, is dropped onto the post from height, or is rocked from side to side to work it down.
- **SOP rule broken:** Steps 3.1, 4.2 and 1.2, lower the part slowly and stop the moment it stops on its own.
- **Coaching note:** let it stop, then let go. A forced part turns a reject into a pass and tells the model to shove.

#### **Violation: Gripper still on the part while the result is read**

- **Visible cue:** the gripper stays closed on the part, rests on it, on a post or on the block, or nudges any of them while the part is being looked at.
- **SOP rule broken:** Steps 1.2, 1.3, 3.2 and 4.3, open the gripper and take it right away before the result is read.
- **Coaching note:** let go, back off, then look. A hand on the part is a hand on the answer.

#### **Violation: Result called before it settles**

- **Visible cue:** a stroke is drawn before the part has been let go and left alone for 2 seconds, or while the part is still moving on the post.
- **SOP rule broken:** Steps 3.2 and 4.3, the result is what the part does once it has been let go and left still for 2 seconds.
- **Coaching note:** wait the two seconds. A part still sliding down the post has not given its answer yet.

#### **Violation: A gauge skipped, or run twice**

- **Visible cue:** a part is binned after only the GO post, a part that stood high on the GO post is not run on the NO GO post, the same post is run again on the same part to check it, or a part goes back on a post after it has been marked.

- **SOP rule broken:** Steps 3 and 4, two gauge runs per part, GO then NO GO, each run once, on every part.
- **Coaching note:** both posts, every part, one run each. A part that already failed still gets the second gauge.

#### **Violation: Part lowered crooked, or resting against a post**

- **Visible cue:** the bore is not lined up with the post as it comes down, the part leans, rocks or hangs on the side of the post, or the part rests on the other post or on the master ring rest instead of sitting square on its own post.
- **SOP rule broken:** Steps 3.1 and 4.2, the bore is lined up with the post and the part comes straight down, square and level, touching nothing but its own post.
- **Coaching note:** square it over the post before you lower it. A crooked part jams high and reads as too small.

#### **Violation: Part rolled, dragged or turned over**

- **Visible cue:** the part is rolled or dragged across the block or the table, tipped onto its side, or rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 2 to 6, every part travels bore up and level, lifted clear of the surface before any horizontal motion.
- **Coaching note:** lift it, carry it level, set it down. A rolled part picks up grit that the gauge then reads.

#### **Violation: Gripper in the bore, or on a post**

- **Visible cue:** the gripper closes on the inside of a part, puts a finger through the bore to carry it, or closes on a gauge post, on the master ring rest or on a bin.
- **SOP rule broken:** Materials checklist and Steps 2 to 6, a part is held on its outside only, and the gripper never touches a post.
- **Coaching note:** the outside is the handle. Anything in the bore or on a post changes what the gauge reads.

#### **Violation: Gauge post or block moved**

- **Visible cue:** a post turns, rocks or lifts in the block, or the gauge block slides, turns or lifts on the table while a part is being lowered or lifted.
- **SOP rule broken:** Materials checklist and Steps 1 to 6, both posts are fixed in the block and the block grips the table.
- **Coaching note:** a post that moves makes every result after it meaningless. Stop and let review flag it.

### **Violation: Master ring binned, tallied, or used as a part**

- **Visible cue:** the master ring gets a tally stroke or a flag mark, goes into a bin, is put in a rack slot, or a part is stood against it or on it to compare the two.
- **SOP rule broken:** Step 1 and Materials checklist, the master ring proves the gauges and is never tallied, never flagged and never binned.
- **Coaching note:** the master is a tool, not a part. It goes back on its rest and nowhere else.

### **Violation: More than one part handled at a time**

- **Visible cue:** the gripper closes on two parts at once, a second part is taken out of the rack before the first is binned, or two parts sit on the block together.
- **SOP rule broken:** Steps 2 to 7, one part at a time, taken, gauged, marked and binned before the next one is touched.
- **Coaching note:** one part, one trip through the steps. A second part on the block loses which stroke belongs to which.

### **Violation: Parts worked out of slot order**

- **Visible cue:** a part is taken from a slot out of turn, or the flag box marked is not the box with that part's slot number on it.
- **SOP rule broken:** Steps 2.1, 5.2 and 7, parts are worked slot 1 through slot 10 in order, and a reject is flagged in the box with its own slot number.
- **Coaching note:** left to right, back row then front row. The box number is the slot number.

### **Violation: Wrong row tallied**

- **Visible cue:** the stroke goes in a row the two gauge results do not say, two rows are marked for one part, two strokes are drawn for one part, or a part is binned with no stroke on the card at all.
- **SOP rule broken:** Step 5.1, one stroke per part, in the GOOD row, the TOO BIG row or the TOO SMALL row, chosen by what the two posts did.
- **Coaching note:** down on GO and high on NO GO is GOOD. Down on NO GO is TOO BIG. High on GO is TOO SMALL. Nothing else.

### **Violation: Reject not flagged, or a good part flagged**

- **Visible cue:** a part tallied TOO BIG or TOO SMALL leaves the block with no X in its flag box, an X is drawn in the flag box of a part tallied GOOD, or the X goes in the box of the wrong slot number.
- **SOP rule broken:** Step 5.2, every reject is flagged in the box with its own slot number and every good part leaves its box blank.
- **Coaching note:** the row says what is wrong, the box says which one. Both, or the card cannot be read.

**Violation: Stroke or mark rubbed out, drawn over, or outside its box**

- **Visible cue:** a stroke or an X is wiped off, drawn over, corrected, or a mark crosses or sits outside a box line, or two tally strokes run into each other so they cannot be counted.
- **SOP rule broken:** Steps 5.1 and 5.2, each mark is one stroke or one X fully inside its own box, and a wrong mark is tagged, never rubbed out.
- **Coaching note:** leave the mistake and let review flag it. A corrected card cannot be counted.

**Violation: Part put in the wrong bin**

- **Visible cue:** a part tallied GOOD goes in the reject bin, a part tallied TOO BIG or TOO SMALL goes in the pass bin, or a part is left on a post or on the table instead of a bin.
- **SOP rule broken:** Step 6.2, the part goes in the bin its own row on the card says.
- **Coaching note:** read the card, then move. The card decides the bin, not the eye.

**Violation: Part taken back out of a bin**

- **Visible cue:** a part is lifted back out of a bin, or a bin is tipped, to move a part that was put in the wrong place.
- **SOP rule broken:** Step 6.2, parts are never taken back out of a bin, and a wrong bin is tagged, not fixed.
- **Coaching note:** once it is in, it stays in. A correction pick makes the episode unlearnable.

**Violation: Part released before it is over its bin**

- **Visible cue:** the gripper opens while the part is still over the block, over open tabletop, or over the wrong bin, or it opens high so the part bounces out.
- **SOP rule broken:** Step 6.2, the gripper opens low over the mouth of the bin the part belongs in.
- **Coaching note:** carry it all the way, low to the rim, then open. A part dropped early is a part on the floor.

**Violation: A part carried over the wrong bin, or over the wrong post**

- **Visible cue:** a good part passes over the reject bin, a reject passes over the pass bin, or a part is carried over the top of the post it is not going onto.
- **SOP rule broken:** Steps 3.1, 4.2 and 6.2, a good part leaves the block to the right, a reject leaves it to the left, and no carry passes over a post it is not going onto.
- **Coaching note:** route around, not across. A part dropped over the wrong thing lands on the wrong thing.

### **Violation: Tool not returned to its home**

- **Visible cue:** the marker is left lying on the card, on the block, in a bin or on the table, the master ring is left on a post or on the table, the gripper opens while the marker is still above its holder, or a tool is still in the gripper when the next step begins.
- **SOP rule broken:** Steps 1.4 and 5.3, the marker returns to its holder and the master ring returns to its rest before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

### **Violation: Spill cleaned up**

- **Visible cue:** after a part lands on the table or on the floor, the gripper is used to sweep, push, or pick it up.
- **SOP rule broken:** Steps 1 to 7, a spill is left where it falls and is cleared in the reset.
- **Coaching note:** cleanup motions are not part of the task. The spill is data, the cleanup is noise.

### **Violation: Bad pick, carry, or release**

- **Visible cue:** an object is dragged along the table with no daylight under it, the gripper shifts or seats its hold again while the object is off the table, a part hangs tilted in the gripper because the grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 6, one object at a time, lifted straight up clear of the surface before any horizontal motion, gripped square on its outside, and tilted no more than a quarter turn.
- **Coaching note:** set it down to change the grip, and lift before the carry. A part that hangs tilted was taken off square.

### **Violation: Placement pushed, or correction limit exceeded**

- **Visible cue:** a part, a bin, the marker, the master ring, the rack or the card is shoved, nudged, or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 7, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

### **Violation: Dropped, collided, or knocked over**

- **Visible cue:** under command, the gripper or a carried object strikes a post, the gauge block, the master ring, the rack, a bin or the card and knocks, tips, slides, or displaces it; or an object falls short of its spot, with no gripper failure, drift, or controller fault visible.
- **SOP rule broken:** Steps 1 to 7, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. The two posts stand in the middle of everything, so the route matters more than the reach.

### **Violation: Manipulation outside the camera frame**

- **Visible cue:** a part pick, a lowering onto a post, a lift off a post, a result or a mark happens partly or wholly outside the environment camera frame, or the arm blocks the camera's view of the foot of a post or of the card while the result is being seen or marked.
- **SOP rule broken:** Steps 1 to 7, every manipulation happens inside the camera frame and both post feet and the card stay visible.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions and keep the arm clear of the posts and the card.

### **Violation: Wrong episode ending**

- **Visible cue:** the episode ends with a part still in the rack or still on a post, a stroke missing or an extra stroke on the card, a reject unflagged, a part in the wrong bin, the master ring off its rest, the marker out of its holder, the arm short of home, or a gripper not fully open.
- **SOP rule broken:** Step 8, confirm the rack, the posts, the bins, the card, the flag strip and the tools, then return the arm home with its gripper open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arm does before the recording ends.

### **Failures that are not violations**

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside the arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective gauge post: one that is worn, nicked, burred or out of size, so the master ring will not go down on the GO post or will not stand high on the NO GO post.

- Defective gauge block: one that slides, turns or lifts on the table under a correct lowering, or whose posts will not stay tight in it.
- Defective shoulder: one that is not flat or not level, so a part that is all the way down still rocks or still shows daylight under one side.
- Defective master ring: one that is chipped, worn or no longer known good.
- Defective rack: one that slides, turns or lifts under a correct pick, or a slot that grips a part so a straight lift will not free it.
- Defective bin: one that will not stand upright in its holder, or that is dented or cracked.
- Dry marker, or a tally card that will not wipe clean so a box or a row starts the episode already marked.
- A tally card printed wrong: a missing row label, flag boxes that touch, flag boxes numbered other than 1 to 10, or a tally box too short to hold ten strokes standing apart.
- A part so close to the size of a post that it neither drops down cleanly nor stands plainly high, so no one can call the result. Replace the part before the next episode.
- Grit, swarf or oil in a bore or on a post that was not there at the start of the episode and turns a good part into a reject.
- Rack holding fewer or more than ten parts at episode start, or the wrong mix of them. Reset error, not a violation.

## **Annotation subtasks (from SOP)**

1. Take the master ring from its rest
2. Lower the master ring onto the GO post
3. Lower the master ring onto the NO GO post
4. Return the master ring to its rest
5. Take one part from its rack slot
6. Lower one part onto the GO post
7. Lift one part off the GO post
8. Lower one part onto the NO GO post
9. Lift one part off the NO GO post
10. Take the marker from its holder
11. Draw one tally stroke on the card
12. Draw one flag mark on the card
13. Return the marker to its holder
14. Drop one part in the pass bin
15. Drop one part in the reject bin
16. Return the arm home and end the episode